Check for updates

Research Article

Received: 2 July 2024 Revised: 26 August 2024 Published online in Wiley Online Library: 18 September 2024

(wileyonlinelibrary.com) DOI 10.1002/ps.8414

Monitoring of plant diseases caused by *Fusarium commune* and *Rhizoctonia solani* in bok choy using hyperspectral remote sensing and machine learning

Derrick Hoang Danh Nguyen,[a*] Arinah Jing Hui Tan,[a] Ronjin Lee,[a] Wei Feng Lim,[a] Jia Yih Wong[a] and Fadhlina Suhaimi[b]

Abstract

BACKGROUND: Local vegetable production is susceptible to various fungal pathogens, the most common and lethal of which are *Fusarium commune* and *Rhizoctonia solani*. Early detection of these pathogens is challenging, and by the time visual symptoms appear, the pathogens may have already spread extensively, causing massive damage to the production. In this study, we explored the use of hyperspectral data for early detection of diseases caused by *F. commune* or *R. solani* in bok choy *Brassica rapa* subsp. *chinensis* by collecting hyperspectral data from healthy plants and plants inoculated with either fungal pathogen in a controlled experimental set-up.

RESULTS: Based on the collected data, we employed various tree-based, distribution-based, geometric, neural networks and ensemble learning algorithms to train detection models. Among the trained models, Multi-Layer Perceptron (MLP) models performed the best with overall accuracy reaching 95.9 ± 0.26%. MLP models could differentiate between healthy and infected plants with 99% precision after 1 day of infection, and distinguish between different fungal pathogens with 99% precision after 2 days. During this period, no visible symptoms of fungal infection could be observed. Further analysis into trained MLP models and general reflectance profiles of plants also revealed a high correlation of the spectral regions 445–460, 560–595, 606–620 and 719–728 nm with fungal infection in bok choy plants.

CONCLUSION: Our findings highlight the potential of hyperspectral imaging as a highly precise early detection tool for fungal diseases in plants.
© 2024 Society of Chemical Industry.

Keywords: remote sensing; hyperspectral imaging; machine learning; plant disease detection

1 INTRODUCTION

Leafy vegetables are among the top agricultural commodities in Singapore, the production of which accounts for 14% of the country local consumption.[1] Owing to Singapore's tropical climate and its high temperature and humidity, local vegetable production is susceptible to various fungal pathogens, the most common and lethal of which are *Fusarium commune* and *Rhizoctonia solani*.[2–4] These fungal pathogens are primarily soil-borne,[5] but can spread quickly in indoor production via irrigation system in local farms. *Fusarium commune* causes *Fusarium* wilt disease,[3,6] where the initial infection of the plant roots leads to vascular blockage, the disruption of water transport and subsequent wilting. Other characteristic symptoms of *Fusarium* wilt include yellowing of leaves, stunting and vascular discoloration.[3] For *R. solani*, infection begins at the plant roots, leading to a basal rot that quickly progresses and spreads to the entire plant.[7] In young seedlings, *R. solani* infection commonly causes damping off, the characteristic softening and collapse of seedling hypocotyls at the soil line.[8]

Given the harmful effects of *F. commune* and *R. solani* on vegetable crop production, detecting these diseases is an inherently challenging task owing to the intricate interactions between host plants and pathogens, as well as the vast array of potential pathogens that can infect the same host species.[9] That means that various different pathogens can induce similar symptoms in host plants; and different hosts within the same species may have variable susceptibility to the same pathogens.[10,11] Additionally, in early stages of infection, infected plants may appear asymptomatic, making visual detection even more challenging.[12]

Considering the challenges in disease monitoring, the potential application of machine vision technologies, particularly hyperspectral imaging (HSI), has been investigated.[13,14] It has been

* *Correspondence to: DHD Nguyen, Plant Science and Health Branch, Horticulture and Community Division, National Parks Board, Singapore. E-mail: derrick_nguyen@nparks.gov.sg*

a *Plant Science and Health Branch, Horticulture and Community Division, National Parks Board, Singapore*

b *Urban Food Solutions Division, Singapore Food Agency, Singapore*

widely shown that when plants are under biotic stress induced by arthropod pests (e.g. spider mites, aphids, thrips) and pathogens (e.g. bacteria, fungi, viruses), these stresses cause changes in leaf reflectance in host plants.[15,16] Such optical manifestation is the result of structural, biochemical and physiological changes in plant cells in response to the infection by fungal pathogens.[9,17–19] Several studies found that differences in reflectance between healthy and infected plants could be detected most consistently around the red-edge (RE) region (700 nm).[20–23] This region, located beyond the visible spectrum detectable by human eyes, grants machine vision technologies such as HSI an edge over conventional visual inspection methods. Besides RE region, various other regions and wavelengths centering around blue (460 nm), green (550 nm) and near infrared (NIR) (800–900 nm) have been shown to be associated with plant biotic stresses.[24–27]

Integrating the mentioned spectral regions, early attempts at disease detection algorithms involved the development of vegetation indices such as the normalized difference vegetation index (NDVI), the water index (WI) and other indices (e.g. EVI), because fungal infections in plants have been observed to be directly correlated with water stress and biochemical changes associated with leaf chlorophyll content.[28–30] In later years, machine learning algorithms such as Discriminant Analysis (DA),[31,32] Support Vector Machines (SVM),[33,34] Random Forests (RF)[35,36] and Logistic Regression (LR)[37,38] have demonstrated high accuracy (>80%) and robustness in disease classification tasks. With the advancement of computer science in the past decade, deep learning approaches such as Multi-Layer Perceptron (MLP), Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) have gained popularity as a result of their power to handle highly dimensional data, an inherent characteristic of hyperspectral images.[39–41] With recent developments in machine learning research, algorithms such as Gaussian Processes classifier (GP), Extreme Gradient Boosting (XGBoost), Attentive Interpretable Tabular Learning (TabNet) and Mixture-of-Experts ensemble learning (MoE) have emerged as new promising approaches in model training for disease detection.[42–44]

Despite the rich body of research in plant disease detection based on HSI data, there are limited studies focusing on *Fusarium* and *Rhizoctonia*. Even among the few published hyperspectral studies on these two pathogens, detection models developed in these studies often have at least one of the following limitations: (i) binary classification (healthy *versus* disease), (ii) strongly symptom-based detection and (iii) lack of temporal assessment of detection. Because different fungal pathogens have different biological characteristics, successful disease management programs require information beyond a binary differentiation between healthy and diseased statuses to devise measures specifically targeting the pathogens infecting host plants. Furthermore, for *F. commune* and *R. solani*, manifestation of signs and symptoms implies a very late stage of infection, by which severe crop loss has been incurred, compromising the operational practicality of symptom-based detection models. Thus, beyond overall detection accuracy, trained models need to be assessed temporally to determine how early into the infection timeline the models can accurately detect.

Considering the discussed limitations and the needs of Singapore local vegetable production, in this study, we aimed to (i) collect hyperspectral data from bok choy *Brassica rapa* subsp. *chinensis* which were either healthy or infected with *Fusarium commune* or *Rhizoctonia solani*, (ii) develop a detection model using various machine learning algorithms to select the best-performing model and (iii) assess performance of the best detection model against different time points.

2 MATERIALS AND METHODS

2.1 Media preparation and use

Potato dextrose agar (PDA; NutriSelect® Plus, Merck, Darmstadt, Germany), aliquoted in 20-mL portions into sterile 9.4-cm-diameter triple-vented disposable plastic petri dishes (Greiner, Kremsmünster, Germany), was used for general fungal growth, whereas wheat bran soil mix (two parts wheat bran, two parts peat substrate and one part water) was used in the preparation of the soil-based inoculum. All media and reverse osmosis (RO) water used were autoclaved at 121 °C for 15 min, with the wheat bran soil mix also being autoclaved twice over the 2 days before use. The wheat bran soil mix itself was autoclaved in 500-g portions in polypropylene bags which have their mouths stoppered with a removable cotton wool plug, and secured with aluminum foil and autoclave tape.

2.2 Preparation of fungal pathogens and inoculum

The *F. commune* K. Skovg., O'Donnell & Nirenberg and *R. solani* Kühn isolates used in this study (P-SPH-2021-12-00013 LCK 74 - 41 and P-SPH-2021-09-00019 PL21-01, respectively) were from the Plant Pathology Laboratory, Plant Science & Health Branch, National Parks Board, Singapore. Isolates of *F. commune* and *R. solani* were incubated at 30 °C in the dark [Fig. 1(B), (C)].

One-week-old cultures of *F. commune* were used to produce a spore suspension by agitating the surface of each fully colonized petri dish with 10 mL sterile RO water. The collected spore suspension was then diluted to a working concentration of 10^{-6} spores mL^{-1} before use as an inoculum. By contrast, the bags of wheat bran soil mix were inoculated with either fungus and were allowed to fully colonize the bags before use.

2.3 Preparation of host plants

Bok choy seeds [*B. rapa* subsp. *chinensis* (L.) Hanelt.] from Singapore Horti® Flora Pte. Ltd were sown into seed plug trays and positioned within a multi-tier seedling rack equipped with TL5 Essential Fluorescent 28 W/865 light tubes (Philips, Singapore). Germinated seedlings were grown under fertilization with 0.1% AB solution for 10 days and transplanted into blue trays, with each tray accommodating 12 plants [Fig. 1(A)]. These young seedlings were then grown under grow light supplied by Ezygro DGTH-25C-CWW LED light tubes. For treatment application, randomized incomplete block design was used with each tray considered as one block, because the soil-borne nature of the involved pathogens made randomized complete block design unfeasible. All 12 seedlings within each tray underwent the same treatment: control or inoculation with *F. commune*, or inoculation with *R. solani*. Each treatment had two trays with a total of 24 plants as replicates. The selection of treatment for each tray was randomized.

2.4 Hyperspectral imaging system

Figure 1(D) shows the HSI system used in this study: this comprised an OCI-F-120 sensor (BaySpec, Inc., San Jose, CA, USA) mounted on a custom-built 2-m elevator scissor-lift mobile robot which can be remotely controlled. The HSI sensor captured 121 wavelengths within the 400–1000 nm range, with a spectral resolution of 5–7 nm.

 © 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

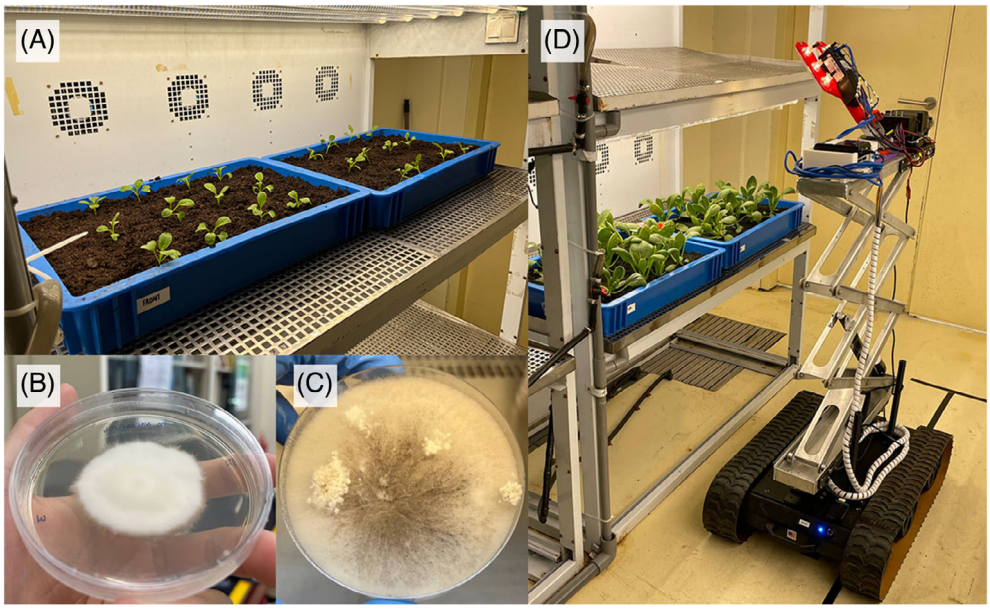


Figure 1. (A) 10-day-old bok choy seedlings arranged in blue trays with Klasmann-Delimann TS1 propagation substrate. (B) Pure culture of *Fusarium commune*. (C) Pure culture of *Rhizoctonia solani*. (D) Hyperspectral imaging system.

Table 1. Overview of collected data. In this study, an independent dataset is defined as data collected from plants grown in the same batch whose seeds were sown on the same day

Class	Number of independent datasets	Number of samples
Control	6	912
Fusarium commune	3	201
Rhizoctonia solani	6	597
Total		1710

For data collection, the HSI camera was positioned at 35° and elevated using the robot to standardize height depending on which tier the plant trays were located [Fig. 1(D)]. Before image acquisition, dark and white calibration were performed by covering the entire field of vision (FOV) of HSI sensor with a cover cap and white standard, respectively. An additional white calibration step was carried out by capturing raw hyperspectral images of plants alongside a white reference Teflon board positioned between trays.

In this study, there were a total of three classes: control bok choy plants with no fungal inoculation, and bok choy plants inoculated with either *F. commune* or *R. solani*, with each class having 24 plants for each trial. Inoculation was performed when bok choy seedlings were transplanted. For the *F. commune* inoculation, the roots of the seedlings were soaked in the *F. commune* spore suspension for 30 min, before transplanting into growing trays, with the roots of each seedling also surrounded by ≈20 mL *F. commune*-inoculated wheat bran soil mix. As *R. solani* does not produce spores, only the latter inoculation step was carried out.

Hyperspectral image data were collected at the following time points: before fungal inoculation, and on 1, 2, 3, 4 and 7 days post-inoculation (dpi). In total six trials were conducted with details shown in Table 1.

2.5 Data processing, spectral filtering and dimension reduction

From raw hyperspectral images, we filtered out background pixels using normalized difference vegetation index (NDVI)[45] to select only plant pixel. In brief, NDVI is calculated using the formula:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

where NIR and Red refer to near-infrared and red light reflectance, respectively.[46] This index was based on the characteristic absorption of plant chlorophyll pigments which absorb most of the incident red light but reflect all wavelengths within the near-infrared region.[47] As a result, plants typically have NDVI values >0.3 whereas other materials have NDVI values <0.3 (e.g. water: −0.5 to −0.1, snow or bare rock: 0 to 0.2, bare soil: 0.1 to <0.3).[47] After filtering with NDVI, reflectance values for each plant pixel were then normalized by dividing by those of white Teflon board and then transformed into a series of point percentage change (PPC) values using the following formula:

$$PPC_n = \frac{RR_{n+1} - RR_{n-1}}{RR_n}$$

where PPC_n is PPC at n^{th} wavelength, RR is relative reflectance, and $n-1$, n and $n+1$ indicate adjacent wavelengths among 121 wavelengths recorded by the HSI sensor. PPC profiles of all pixels for a plant were then averaged to obtain one PPC profile to represent the entire plant. The whole process is visualized in Fig. 2.

2.6 Classification model training and evaluation

All machine learning implementation and data analysis presented in this article were performed using Keras API from Tensorflow v2.13.0 on Google Colab Pro. Across various machine learning studies, common train/test split ratios used include 50/50, 70/30, 80/20 and 90/10.[48–50] Although the optimal train/test split ratio

 © 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

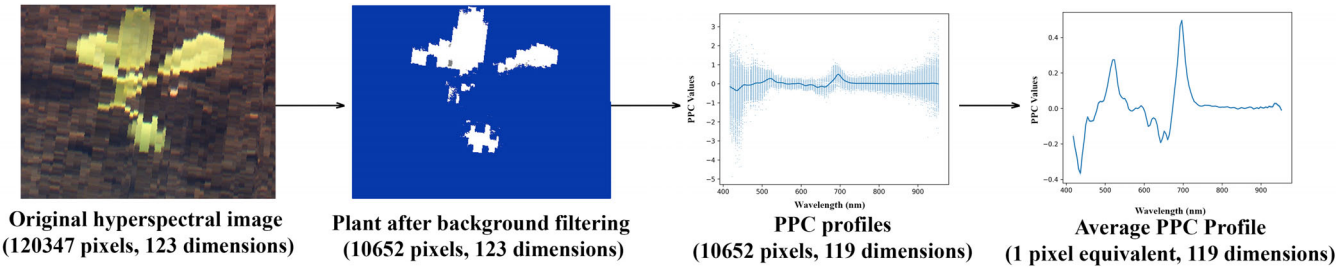


Figure 2. Spectral filtering and Point Percentage Change (PPC) transformation on an example hyperspectral image that contains 120 347 pixels with 123 dimensions (two dimensions for pixel 2D positions and 121 dimensions for reflectance values at 121 wavelengths recorded by hyperspectral sensor).

Table 2. Summary of machine learning algorithms used and their corresponding optimized parameters to obtain highest average accuracy in 10-fold cross-validation. Optimized parameters were obtained via grid search method for those with numerical values

Model type	Model	Parameters
Tree-based	Random Forest (RF)	Max depth = 20, Min samples leaf = 1, Min sample split = 2, No of estimators = 500
	Extreme Gradient Boosting (XGBoost)	Learning rate = 0.1, Max depth = 5, No of estimators = 200
Distribution-based	Logistic Regression (LR)	C = 2000, Max iteration = 1000, Solver = lbfgs
	Quadratic Discriminant Analysis (QDA)	Tolerance for convergence = 10^{-6}, Class priors = [0.3,0.4,0.3]
	Gaussian Processes (GP)	Kernel = RBF (0.1), No of random restarts = 2, Optimizer = fmin_l_bfgs_b
Geometric	Support Vector Machine (SVM)	Kernal = RBF, C = 100, Gamma = scale
Neural Network	Multi-Layer Perceptron (MLP)	Architecture: [Dense (512), Dense (128), Dense (32)], Dropout = 0.2, Activation = ReLU, Solver = Adam, Max epoch = 100, Batch size = 32, Learning rate = 0.001
	Attentive Interpretable Tabular Learning (TabNet)	Model dimension = 32, Attention dimension = 16, No. of steps = 5, Gamma = 1, Lambda sparse = 0.001, Max epoch = 150, Batch size = 32
	Long Short-Term Memory (LSTM)	Architecture: [LSTM (512), LSTM (256), Dense (16)], Dropout = 0.3, Activation = ReLU, Solver = Adam, Max epoch = 200, Batch size = 32, Learning rate = 0.001
Ensemble Learning	Mixture of Experts (MoE)	No of experts = 5, Hidden dimension size = 128, No of top gates = 3, Learning rate = 0.01, Max epoch = 200

varies with the nature and size of original dataset and machine learning objectives, the consensus is that split ratio should allow maximized usage of data for training yet maintain a substantial test dataset for evaluation of trained model. From the original dataset of 1710 samples, we found that a 90/10 train/test split allowed us to use most of the data for training and maintained a sizeable test dataset with 171 samples.

For the training portion of the data, we fitted various machine learning algorithms and fine-tuned relevant parameters such that trained models for a particular algorithm achieved the optimal average accuracy for 10-fold cross-validations. The details of machine learning algorithms and their tuned parameters are summarized in Table 2. For each algorithm, trained models were fitted into test data to obtain classification accuracy, precision, recall and F1 scores. The best algorithm, defined as one with the highest average accuracy in 10-fold cross-validation, was selected for further model visualization described in a subsequent section.

2.7 Model visualization

In order to visualize trained models, permutation importance[50] and Shapley additive explanations (SHAP) feature importance[51] were used. In this study, the implementation of permutation importance involved randomly shuffling four times PPC values at one wavelength while keeping the rest of PPC values unchanged. For each shuffling event, the trained model was fitted to the shuffled test data to obtain change in accuracy as compared to when the same model was fitted to unshuffled test data. Changes in accuracy were then averaged across four shuffling events. The same process was applied to the rest of the 119 wavelengths for all 10 models trained from cross-validation. For SHAP feature importance, the entire test dataset was fitted to model with the highest accuracy using explainer.shap_values() from the SHAP package. The top 10 wavelengths with the highest influence on model outputs were determined.

3 RESULTS

3.1 Manifestation of symptoms of infestation and corresponding PPC values

For *F. commune*, infected plants displayed no visible symptoms within 1 dpi. At 2 dpi, very mild signs of wilting started to appear in the younger leaves which can be easily mistaken for water stress or completely overlooked through visual inspection [Fig. 3(A)]. This wilt quickly spread to entire plants by 3 dpi, with chlorosis and brown blotches being clearly visible, especially near the major leaf veins [Fig. 3(B)]. By the 7 dpi, entire plants had completely wilted [Fig. 3(C)] and ≈75% of the plants had died [Fig. 3(D)].

Meanwhile, no visible symptoms were observed within 2 dpi with *R. solani*. Symptoms of damping-off became visible on 3 dpi as bases of young seedlings started to rot, leading to plant

© 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



Figure 3. Appearance of plants infested by *Fusarium commune* (A–D) and *Rhizoctonia solani* (E–H) over time postinoculation with fungus: (A) 2, (B) 3, (C, D) 7, (E) 3, (F) 4 and (G, H) 7 dpi.

collapse [Fig. 3(E)]. The rot spread quickly to older leaves by 4 dpi [Fig. 3(F)] and entire plants by 7 dpi [Fig. 3(G)]. Although the stunting effect could be observed clearly by 7 dpi, the mortality rate was found to be ≈20% for plants infected with *R. solani* [Fig. 3(H)].

Figure 4 shows the average PPC profiles of control plants and plants inoculated with either *F. commune* or *R. solani*. Significant separation in PPC profiles between all three classes could be observed in the 719–724 nm range 1 dpi. This separation expanded to a wider spectral region of 701–728 nm by 2 dpi, remained significant for the rest of the trial and became more prominent over time. By 7 dpi, significant separation in PPC profiles could be observed in the 560–595 and 606–620 nm ranges.

3.2 Model performance

Table 3(A) summarizes overall model accuracy, precision, recall and F1 scores for various machine learning algorithms. Except for GP models, tree-based and distribution-based models had inferior classification accuracy compared to that from geometric, neural network and ensemble-learning models. MLP provided best results with the highest scores for all metrics while having comparatively fast training time. Therefore, we obtained the confusion matrix of the MLP model with the highest overall classification accuracy [Table 3(B)] and visualized the feature importance of the MLP models via permutation importance and SHAP analysis in Section 3.2.

© 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

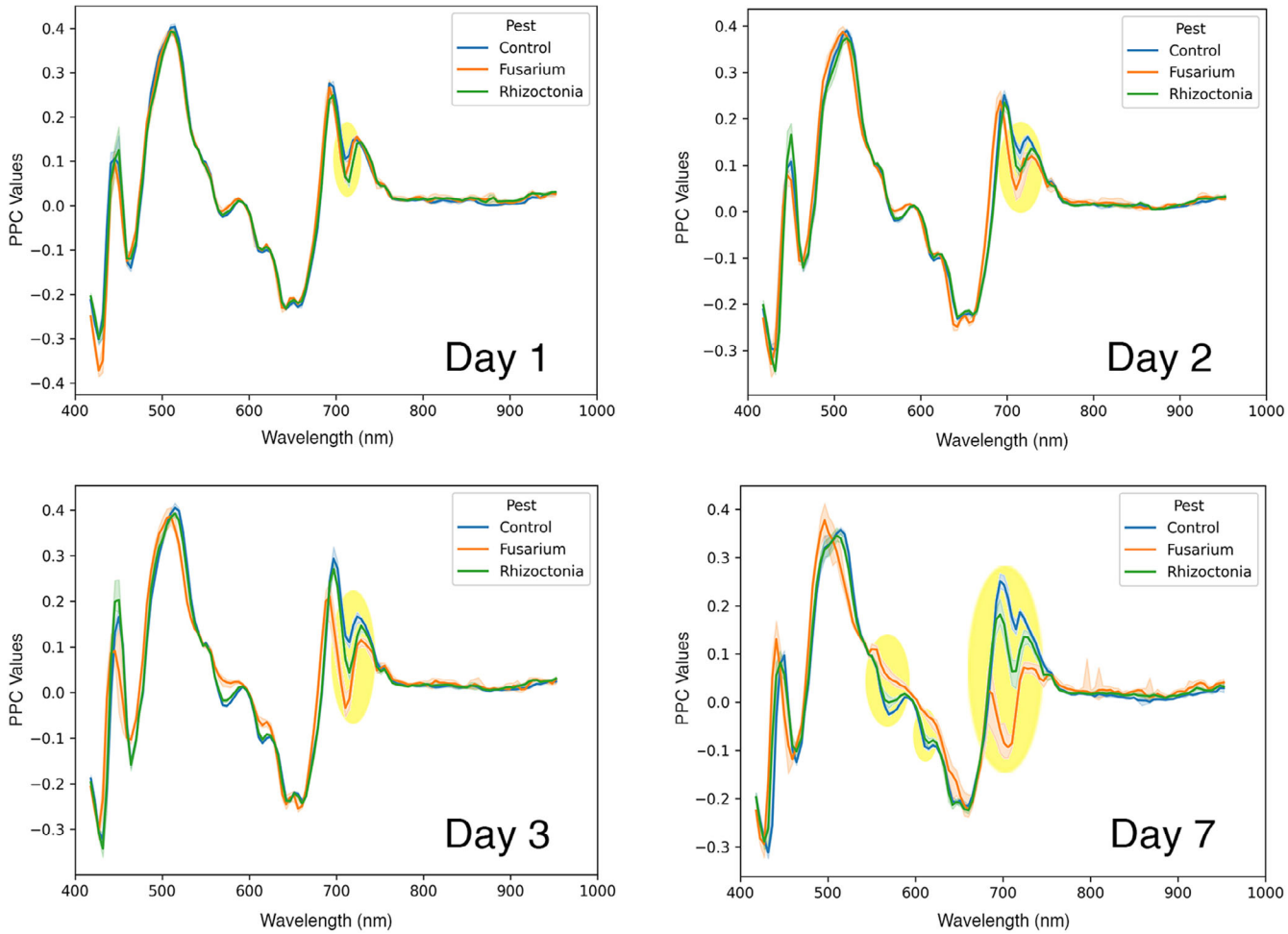


Figure 4. Average Point Percentage Change (PPC) values with 95% confidence interval for different classes across different time points after fungal inoculation. Spectral regions where statistically significant separations in PPC values between different classes can be observed are highlighted in yellow.

In terms of temporal variation, trained MLP models could detect fungal infection at 1 dpi with average 'Control' class precision ≤99% (Fig. 5). By 2 dpi, precision in predicting the type of fungal pathogens reached an average value of 99% for both fungal classes. Classification precision classes remained relatively high for the rest of the trials, except for *R. solani* on 3 dpi.

3.3 Model visualization

The reduction in model accuracy when PPC values of any wavelength were randomly shuffled ranged from +2.9% to −12.3% (Fig. 6). The 719–728 nm (RE region) was the most important to trained MLP models with the highest accuracy drop observed from the permutation important heat map. The spectral region 430–440 nm (blue region) was moderately important to trained MLP models with accuracy dropping by ≈3–5% when PPCs within this region were shuffled.

As seen from Fig. 7, when the best MLP model was fitted into the test dataset, wavelength 719 nm (RE region) was found to be among the top 10 wavelengths with the most impact on model output for all three classes. Wavelengths 445and 464 nm (blue region) were unique to fungal infection whereas high PPC values for wavelengths 440(blue region), 560(green region), 710and 724 nm (RE region) were unique signatures of control plants.

4 DISCUSSION

4.1 Machine learning algorithm performance for tabular data

In this study, we experimented with an array of machine learning methods. Based on performance metrics such as accuracy, precision, recall and F1 scores, it was evident that geometric, neural network and ensemble learning generally provided far more desirable results than distribution-based and tree-based algorithms [Table 3(A)]. The presented result, in a way, was unexpected because the XGBoost algorithm has been considered the go-to learning algorithm for tabular data in recent years for its superior performance over other algorithms.[52,53] Various studies have compared the performance of XGBoost and other machine learning algorithms, especially neural networks, on different tabular datasets, finding that XGBoost models often outperformed the rest in terms of training time and accuracy.[52,54,55] In our study, hyperspectral images after spectral filtering and dimension reduction processes produced a tabular dataset with each plant represented by a string of PPC values across 119 wavelengths. However, the average accuracy of XGBoost models was found to be among the lowest for our dataset. Further analysis of published studies revealed that cases in which XGBoost results were better often involved discrete independent input features, whereas neural network models tended to provide better results for cases in

 © 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3(A). Average model training time per fold, accuracy, precision, recall and F1 scores with standard error for trained models obtained from 10-fold cross-validation

Model type	Model	Average training time per fold ± SE (seconds)	Average accuracy ± SE (%)	Average precision ± SE (%)	Average recall ± SE (%)	Average F1 scores ± SE (%)
Tree-based	Random Forest (RF)	4.82 ± 0.01	82.69 ± 0.36	83.14 ± 0.47	77.67 ± 0.50	79.78 ± 0.49
	Extreme Gradient Boosting (XGBoost)	1.29 ± 0.01	87.48 ± 0.40	82.26 ± 0.34	85.69 ± 0.45	86.24 ± 0.34
Distribution-based	Logistic Regression (LR)	0.60 ± 0.01	86.9 ± 0.43	86.34 ± 0.60	86.12 ± 0.60	86.11 ± 0.56
	Quadratic Discriminant Analysis (QDA)	0.044 ± 0.001	82.34 ± 0.39	86.93 ± 0.26	69.81 ± 0.64	70.00 ± 0.88
	Gaussian Processes (GP)	154.9 ± 6.7	93.62 ± 1.29	93.86 ± 0.79	91.26 ± 2.59	91.18 ± 2.77
Geometric	Support Vector Machine (SVM)	0.110 ± 0.001	94.50 ± 0.15	94.41 ± 0.14	93.84 ± 0.22	93.96 ± 0.17
Neural Network	**Multi-Layer Perceptron (MLP)**	**21.77 ± 0.12**	**95.90 ± 0.26**	**95.91 ± 0.35**	**95.98 ± 0.35**	**95.88 ± 0.29**
	Attentive Interpretable Tabular Learning (TabNet)	355.8 ± 4.1	94.15 ± 0.62	93.58 ± 0.75	93.65 ± 0.62	93.5 ± 0.66
	Long Short-Term Memory (LSTM)	123.4 ± 0.5	94.15 ± 0.57	93.18 ± 0.43	94.14 ± 0.83	93.4 ± 0.60
Ensemble Learning	Mixture of Experts (MoE)	32.33 ± 0.25	95.03 ± 0.31	93.71 ± 0.40	94.93 ± 0.34	94.18 ± 0.35

Table 3(B). Confusion matrix of best Multi-Layer Perceptron (MLP) model when fitted to test data

		Predicted			
		Control	*Fusarium commune*	*Rhizoctonia solani*	Recall (%)
Actual	Control	88	1	0	98.88
	Fusarium commune	0	27	0	100.00
	Rhizoctonia solani	2	1	52	94.55
Precision (%)		97.78	93.10	100	Overall accuracy: 97.66

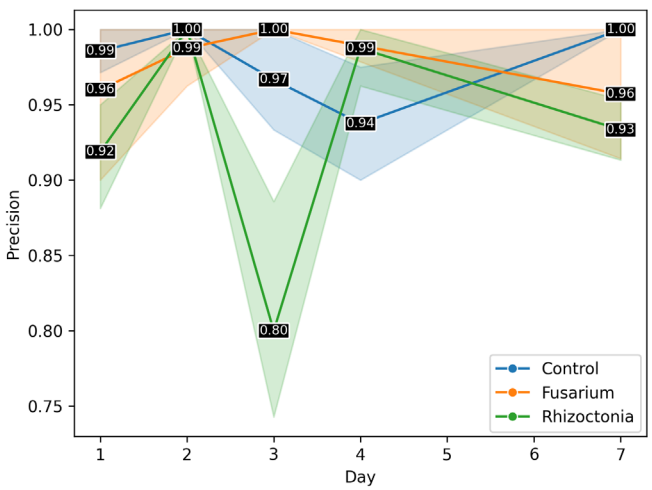


Figure 5. Average prediction accuracy with 95% confidence intervals when 10 trained Multi-Layer Perceptron (MLP) models obtained from 10-fold cross-validation were fitted into the test dataset for different classes over different time points of infestation.

which input features were continuous and not completely independent.[55–58] Being a tree-based model, the XGBoost algorithm relies on recursively partitioning the feature space into regions and learns a series of hierarchical if-else rules to form predictions.[59] Therefore, we surmise that feature space partition is relatively more challenging for continuous and collinear features, characteristics of hyperspectral data, as the boundaries between partitions are not as clear as those observed in discrete and independent features.

Furthermore, among the neural network architectures experimented in this study, MLP models outperformed the TabNet and LSTM models in all tracked metrics while maintaining relatively quick training time [Table 3(A)]. Training convergence for MLP models also happened earlier than TabNet and LSTM as seen from the optimized maximum number of epochs (Table 2). Such results highlight that since its introduction by Kingma and Ba,[60] the current version of Adaptative Moment Estimation algorithm, also known as Adam, has become such a powerful deep learning optimizer that simple feed-forward architectures such as those seen in MLP and TabNet can still achieve comparable or even better results than more complex architectures as observed in LSTM.

© 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

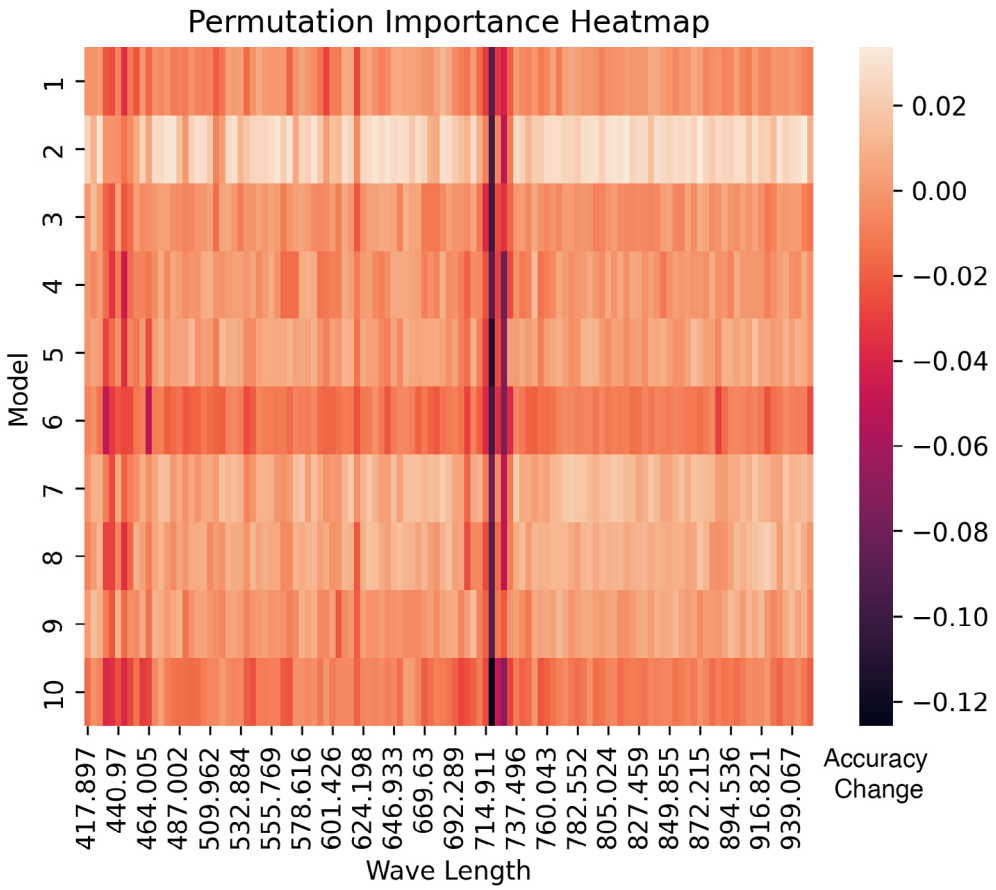


Figure 6. Average prediction accuracy changes for 10 trained Multi-Layer Perceptron (MLP) models when individual wavelength was shuffled four times as part of a permutation importance procedure. Greater accuracy reduction indicates higher importance of wavelength for trained MLP models.

4.2 Spectral reflectance associated with fungal diseases and early detection capacity of hyperspectral-based model

As seen from average PPC profiles across three classes over time, spectral region 719–728 nm located in RE produced significant separation in PPC values between classes as early as 1 dpi and remained consistently associated with fungal infection throughout all trials (Fig. 4). All 10 trained MLP models also recognized the importance of this spectral region as the models were found to rely more heavily on this region as revealed by permutation importance (Fig. 6) and SHAP (Fig. 7) analyses. Additionally, 719 nm appeared to be the most critical wavelength in multi-class detection, whereas 445 and 464 nm seemed associated with fungal infection in bok choy plants (Fig. 7). The great importance of RE regions has been widely reported to be linked to plant stresses in various HSI studies.[61–63] The association between RE reflectance and fungal diseases in plants is likely to be related to plant self-defense mechanisms in response to infection which results in release and/or biosynthesis of secondary metabolites, leading to changes in the way infected plant cells absorb and reflect light.[64] In bok choy, broad-spectrum antifungal defense has been shown to be mediated by glucosinolate[65] and the concentration of this molecule has been found to be correlated with RE reflectance in plant cells.[66]

More importantly, besides identifying discriminative wavelengths for fungal disease detection, top performing models in our study could differentiate between different fungal pathogens accurately [Table 3(A)]. Additionally, prediction precision values could reach 99%, 96% and 92% for control plants and plants inoculated with *F. commune* and *R. solani*, respectively, just after 1 dpi (Fig. 5). The precision values reached ≥99% or all three classes 2 dpi. Such results indicate the great advantage of HSI as detection could be achieved with high accuracy within the first 2 days. During this period, as formerly discussed in the fungal disease manifestation section, there were no visible signs and symptoms for both *F. commune* and *R. solani*.

4.3 Limitations

Despite the good performances of our trained model in the detection of fungal diseases, we recognized two main limitations in our study. First, the training dataset was imbalanced with control plants accounting for more than half and *F. commune* class only accounting for ~11% of the dataset. Such data imbalance was unavoidable as a consequence of the different biological characteristics of these fungal pathogens, leading to difficulties in aligning *F. commune* and *R. solani* fungal cultures for experiments. Additionally, not all plants inoculated with fungal pathogens survived towards the end of every trial, leading to the lower number of samples for *F. commune* and *R. solani* classes as compared to the control. From a model training perspective, machine learning algorithms are known to be quite sensitive to data imbalance and tend to develop bias towards majority class.[67] To assess the severity of data imbalance on model robustness, metrics including precision, recall and F1 scores can be used.[68] As seen from Table 3(A), there were no significant differences between the mentioned metrics regardless of the type of model used. A closer look into

© 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

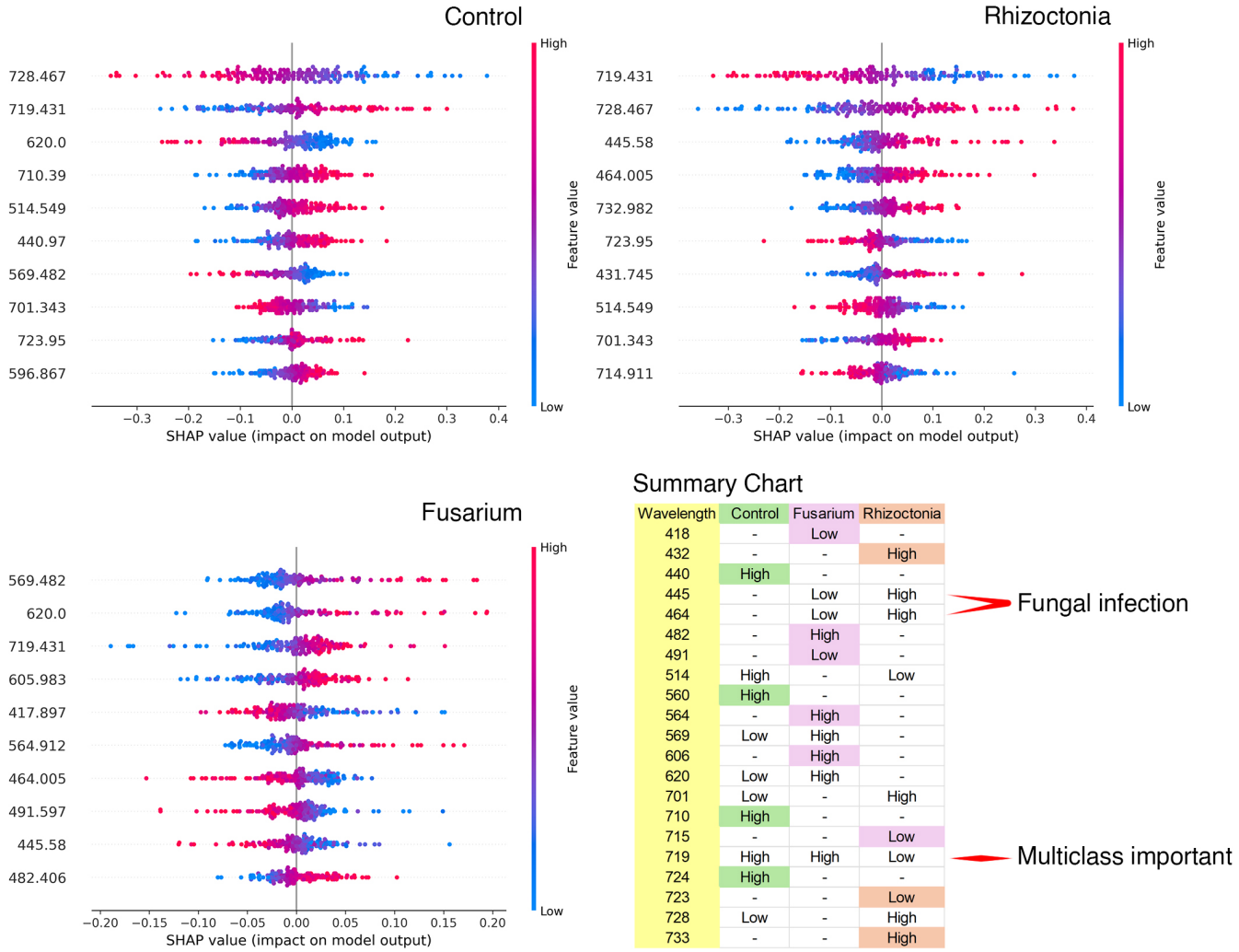


Figure 7. Shapley additive explanations (SHAP) values and distribution for the top 10 wavelengths with the greatest impact on model output for each class. For quick reference, a summary chart was developed to show wavelengths with the most impact on model output across three classes. High and Low indicate the magnitude of feature value that would increase the probability of model to classifying a plant to a particular class. '-' indicates that the wavelength is not among the top 10 wavelengths for a class. Highlighted cells with corresponding color indicate that the wavelength is unique to a class.

the confusion matrix of the best MLP model for the test dataset [Table 3(B)] indicated no apparent bias towards majority class. Thus, despite the existence of imbalance in the training dataset, its effect on robustness of trained models was minimal.

The second limitation which we noticed in our current study is the poor performance of trained models in detecting plants inoculated with *R. solani* on 3 dpi. The sudden drop in precision to ≈80% on 3 dpi was rather an anomaly as precisions for this class were >90% for the remaining time points. This result could be to the consequence of an unknown biochemical change in plants happening on 3 dpi and the limitation of training algorithms used. Regardless, from a practical perspective, it is more important to be able to detect fungal infection within 2 dpi because visible damping-off symptoms caused by *R. solani* could be observed from 3 dpi.

5 CONCLUSION

In this study, we found that neural network, geometric and ensemble learning algorithms were effective in developing accurate prediction models for detection of fungal diseases. We demonstrated that MLP models trained on HSI data could detect fungal diseases and differentiate between different fungal pathogens accurately as early as 1 dpi, before visible signs and symptoms could be observed. Our future work would involve expanding the training dataset to include more fungal pathogens, especially foliar species, and performing more fine-tuning to enhance the multiclass detection capacity of trained models.

ACKNOWLEDGEMENTS

This study was partly sponsored through Innovation Funding by Singapore Food Agency (SFA). The authors are grateful to Dr. Quang Minh Hoang for his invaluable review of this article.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

 © 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

CONFLICT OF INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

1 SFA, Our Singapore Food Story: Grow Local. https://www.sfa.gov.sg/food-farming/sgfoodstory/grow-local [accessed 2023].
2 Okungbowa F and Shittu H, Fusarium wilts: an overview. *Environ Res J* **6**:83–102 (2012).
3 Hua GKH, Bertier L, Soltaninejad S and Höfte M, Cropping systems and cultural practices determine the Rhizoctonia anastomosis groups associated with Brassica spp. in Vietnam. *PLoS One* **9**:e111750 (2014).
4 Rodriguez MH, Bandte M, Gaskin T, Fischer G and Büttner C, Efficacy of electrolytically-derived disinfectant against dispersal of Fusarium oxysporum and Rhizoctonia solani in hydroponic tomatoes. *Sci Hortic* **234**:116–125 (2018).
5 Gordon TR, Fusarium oxysporum and the Fusarium wilt syndrome. *Ann Rev Phytopathol* **55**:23–39 (2017).
6 Ajayi-Oyetunde O and Bradley C, Rhizoctonia solani: taxonomy, population biology and management of rhizoctonia seedling disease of soybean. *Plant Pathol* **67**:3–17 (2018).
7 Kataria H and Verma P, Rhizoctonia solani damping-off and root rot in oilseed rape and canola. *Crop Prot* **11**:8–13 (1992).
8 Mahlein A-K, Kuska MT, Behmann J, Polder G and Walter A, Hyperspectral sensors and imaging technologies in phytopathology: state of the art. *Ann Review Phytopathol* **56**:535–558 (2018).
9 Martinelli F, Scalenghe R, Davino S, Panno S, Scuderi G, Ruisi P *et al.*, Advanced methods of plant disease detection. A review. *Agron Sustainable Dev* **35**:1–25 (2015).
10 Sandhu GK and Kaur R, Plant disease detection techniques: a review, in *2019 International Conference on Automation, Computational and Technology Management (ICACTM)*. IEEE, pp. 34–38 (2019).
11 Koike S, Subbarao K, Davis RM and Turini T, *Vegetable Diseases Caused by Soilborne Pathogens*. UCANR Publications, (2003).
12 Calvini R, Ulrici A and Amigo JM, Growing applications of hyperspectral and multispectral imaging. *Data Handl Sci Technol* **32**:605–629 (2019).
13 Abd El-Ghany NM, Abd El-Aziz SE and Marei SS, A review: application of remote sensing as a promising strategy for insect pests and diseases management. *Environ Sci Pollut Res* **27**:33503–33515 (2020).
14 Karuppiah K, Senthilkumar T, Jayas D and White N, Detection of fungal infection in five different pulses using near-infrared hyperspectral imaging. *J Stored Prod Res* **65**:13–18 (2016).
15 Nguyen D, Tan A, Lee R, Lim WF, Hui TF and Suhaimi F, Early detection of infestation by mustard aphid, vegetable thrips and two-spotted spider mite in bok choy with deep neural network (DNN) classification model using hyperspectral imaging data. *Comput Electron Agric* **220**:108892 (2024).
16 Hatfield P and Pinter P Jr, Remote sensing for crop protection. *Crop Prot* **12**:403–413 (1993).
17 Riley J, Remote sensing in entomology. *Annu Rev Entomol* **34**:247–271 (1989).
18 Moran MS, Inoue Y and Barnes E, Opportunities and limitations for image-based remote sensing in precision crop management. *Remote Sens Environ* **61**:319–346 (1997).
19 Fernández CI, Leblon B, Haddadi A, Wang K and Wang J, Potato late blight detection at the leaf and canopy levels based in the red and red-edge spectral regions. *Remote Sensing* **12**:1292 (2020).
20 Hu X, Qiao H, Chen B and Si H, A novel approach to grade cotton aphid damage severity with hyperspectral index reconstruction. *Appl Sci* **12**:8760 (2022).
21 Vanegas F, Bratanov D, Powell K, Weiss J and Gonzalez F, A novel methodology for improving plant pest surveillance in vineyards and crops using UAV-based hyperspectral and spatial data. *Sensors* **18**:260 (2018).
22 Jiang X, Zhen J, Miao J, Zhao D, Shen Z, Jiang J *et al.*, Newly-developed three-band hyperspectral vegetation index for estimating leaf relative chlorophyll content of mangrove under different severities of pest and disease. *Ecol Indic* **140**:108978 (2022).
23 Zhao Y, Yu K, Feng C, Cen H and He Y, Early detection of aphid (Myzus persicae) infestation on Chinese cabbage by hyperspectral imaging and feature extraction. *Trans ASABE* **60**:1045–1051 (2017).
24 Kumar J, Vashisth A, Sehgal V and Gupta V, Identification of aphid infestation in mustard by hyperspectral remote sensing. *J Agric Phys* **10**:53–60 (2010).
25 Ranjitha G, Srinivasan M and Rajesh A, Detection and estimation of damage caused by thrips Thrips tabaci (Lind) of cotton using hyperspectral radiometer. *Agrotechnology* **3**:123 (2014).
26 Nansen C, Use of variogram parameters in analysis of hyperspectral imaging data acquired from dual-stressed crop leaves. *Remote Sens (Basel)* **4**:180–193 (2012).
27 Ashourloo D, Aghighi H, Matkan AA, Mobasheri MR and Rad AM, An investigation into machine learning regression techniques for the leaf rust disease detection using hyperspectral measurement. *IEEE J Sel Top Appl Earth Obs Remote Sens* **9**:4344–4351 (2016).
28 Feng W, Shen W, He L, Duan J, Guo B, Li Y *et al.*, Improved remote sensing detection of wheat powdery mildew using dual-green vegetation indices. *Precis Agric* **17**:608–627 (2016).
29 Barreto A, Paulus S, Varrelmann M and Mahlein A-K, Hyperspectral imaging of symptoms induced by Rhizoctonia solani in sugar beet: comparison of input data and different machine learning algorithms. *J Plant Dis Prot* **127**:441–451 (2020).
30 Pydipati R, Burks T and Lee W, Identification of citrus disease using color texture features and discriminant analysis. *Comput Electron Agric* **52**:49–59 (2006).
31 Shi Y, Huang W, Luo J, Huang L and Zhou X, Detection and discrimination of pests and diseases in winter wheat based on spectral indices and kernel discriminant analysis. *Comput Electron Agric* **141**:171–180 (2017).
32 Kaur R and Kang SS, An enhancement in classifier support vector machine to improve plant disease detection, in *2015 IEEE 3rd International Conference on MOOCs, Innovation and Technology in Education (MITE)*. IEEE, pp. 135–140 (2015).
33 Hussein MA and Abbas AH, Plant leaf disease detection using support vector machine. *Al-Mustansiriyah J Sci* **30**:105–110 (2019).
34 Govardhan M and Veena M, Diagnosis of tomato plant diseases using random forest, in *2019 Global Conference for Advancement in Technology (GCAT)*, IEEE, 1–5 (2019).
35 Hatuwal BK, Shakya A and Joshi B, Plant leaf disease recognition using random forest, KNN, SVM and CNN. *Polibits* **62**:13–19 (2020).
36 Harikrishnan R and Río L, A logistic regression model for predicting risk of white mold incidence on dry bean in North Dakota. *Plant Dis* **92**:42–46 (2008).
37 Rajeshwari T, Vardhini PH, Reddy KMK, Priya KK and Sreeja K, Smart agriculture implementation using IoT and leaf disease detection using logistic regression, in *2021 4th International Conference on Recent Developments in Control, Automation & Power Engineering (RDCAPE)*. IEEE, pp. 619–623 (2021).
38 Jin X, Jie L, Wang S, Qi HJ and Li SW, Classifying wheat hyperspectral pixels of healthy heads and Fusarium head blight disease using a deep neural network in the wild field. *Remote Sens (Basel)* **10**:395 (2018).
39 Golhani K, Balasundram SK, Vadamalai G and Pradhan B, A review of neural networks in plant disease detection using hyperspectral data. *Inf Process Agric* **5**:354–371 (2018).
40 Weng S, Han K, Chu Z, Zhu G, Liu C, Zhu Z *et al.*, Reflectance images of effective wavelengths from hyperspectral imaging for identification of Fusarium head blight-infected wheat kernels combined with a residual attention convolution neural network. *Comput Electron Agric* **190**:106483 (2021).
41 Reis Pereira M, Verrelst J, Tosin R, Rivera Caicedo JP, Tavares F, Neves dos Santos F *et al.*, Plant disease diagnosis based on hyperspectral sensing: comparative analysis of parametric spectral vegetation indices and nonparametric gaussian process classification approaches. *Agronomy* **14**:493 (2024).
42 Huang L, Liu Y, Huang W, Dong Y, Ma H, Wu K *et al.*, Combining random forest and XGBoost methods in detecting early and mid-term winter wheat stripe rust using canopy level hyperspectral measurements. *Agriculture* **12**:74 (2022).
43 Shah C, Du Q and Xu Y, Enhanced TabNet: attentive interpretable tabular learning for hyperspectral image classification. *Remote Sens* **14**:716 (2022).
44 Rouse Jr JW, Haas R, Schell J and Deering D, *Monitoring Vegetation Systems in the great plains with ERTS*. (1974).
45 Huang S, Tang L, Hupy JP, Wang Y and Shao G, A commentary review on the use of normalized difference vegetation index (NDVI) in the era of popular remote sensing. *J For Res* **32**:1–6 (2021).

© 2024 Society of Chemical Industry.

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

46 Jones HG and Vaughan RA, *Remote Sensing of Vegetation: Principles, Techniques, and Applications*. Oxford University Press, USA (2010).
47 Muraina I, Ideal dataset splitting ratios in machine learning algorithms: general concerns for data scientists and data analysts, in *7th International Mardin Artuklu Scientific Research Conference*, pp. 496–504 (2022).
48 Singh V, Pencina M, Einstein AJ, Liang JX, Berman DS and Slomka P, Impact of train/test sample regimen on performance estimate stability of machine learning in cardiovascular imaging. *Sci Rep* **11**:14490 (2021).
49 Rácz A, Bajusz D and Héberger K, Effect of dataset size and train/test split ratios in QSAR/QSPR multiclass classification. *Molecules* **26**: 1111 (2021).
50 Altmann A, Toloşi L, Sander O and Lengauer T, Permutation importance: a corrected feature importance measure. *Bioinformatics* **26**: 1340–1347 (2010).
51 Misheva BH, Osterrieder J, Hirsa A, Kulkarni O and Lin SF, Explainable AI in credit risk management. *arXiv preprint arXiv:210300949* 2021).
52 Shwartz-Ziv R and Armon A, Tabular data: deep learning is not all you need. *Inf Fusion* **81**:84–90 (2022).
53 Hwang Y and Song J, Recent deep learning methods for tabular data. *Commun Stat Appl Methods* **30**:215–226 (2023).
54 Fayaz SA, Zaman M, Kaul S and Butt MA, Is deep learning on tabular data enough? An assessment. *Int J Adv Comput Sci Appl* **13**:466–473 (2022).
55 Borisov V, Leemann T, Seßler K, Haug J, Pawelczyk M and Kasneci G, Deep neural networks and tabular data: a survey. *IEEE Trans Neural Networks Learn Systems* **35**:7499–7519 (2022).
56 Gorishniy Y, Rubachev I, Khrulkov V and Babenko A, Revisiting deep learning models for tabular data. *Adv Neural Inf Process Syst* **34**: 18932–18943 (2021).
57 Gorishniy Y, Rubachev I and Babenko A, On embeddings for numerical features in tabular deep learning. *Adv Neural Inf Process Syst* **35**: 24991–25004 (2022).
58 Nandita D, Ljubomir B and Inflammatix I, Searching for the Best Tabular Data Classifiers. https://inflammatix.com/searching-for-the-best-tabular-data-classifiers-blog/ [accessed 2023].
59 Azmi SS and Baliga S, An overview of boosting decision tree algorithms utilizing AdaBoost and XGBoost boosting strategies. *Int Res J Eng Technol* **7**:6867–6870 (2020).
60 Kingma DP and Ba J, Adam: A method for stochastic optimization. *arXiv preprint arXiv:14126980* 2014).
61 Teena M, Manickavasagan A, Ravikanth L and Jayas D, Near infrared (NIR) hyperspectral imaging to classify fungal infected date fruits. *J Stored Prod Res* **59**:306–313 (2014).
62 Zhao Y-R, Yu K-Q, Li X and He Y, Detection of fungus infection on petals of rapeseed (Brassica napus L.) using NIR hyperspectral imaging. *Sci Rep* **6**:38878 (2016).
63 Moghadam P, Ward D, Goan E, Jayawardena S, Sikka P and Hernandez E, Plant disease detection using hyperspectral imaging, in *2017 International Conference on Digital Image Computing: Techniques and Applications (DICTA)*. IEEE, pp. 1–8 (2017).
64 Terentev A, Dolzhenko V, Fedotov A and Eremenko D, Current state of hyperspectral remote sensing for early plant disease detection: a review. *Sensors* **22**:757 (2022).
65 Bednarek P, Piślewska-Bednarek M, Svatoš A, Schneider B, Doubský J, Mansurova M *et al.*, A glucosinolate metabolism pathway in living plant cells mediates broad-spectrum antifungal defense. *Science* **323**:101–106 (2009).
66 Velasco L and Becker H, Analysis of total glucosinolate content and individual glucosinolates in Brassica spp. by near-infrared reflectance spectroscopy. *Plant Breed* **117**:97–102 (1998).
67 Krawczyk B, Learning from imbalanced data: open challenges and future directions. *Prog Artif Intell* **5**:221–232 (2016).
68 Huang CY and Dai HL, Learning from class-imbalanced data: review of data driven methods and algorithm driven methods. *Data Sci Finance Econ* **1**:21–36 (2021).

15264998, 2025, 1, Downloaded from https://scijournals.onlinelibrary.wiley.com/doi/10.1002/ps.8414 by Northwest Agri & Forestry, Wiley Online Library on [07/09/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License